

This document provides a **step-by-step guide** to setting up the **ELK Stack (Elasticsearch, Logstash, Kibana)** with **Filebeat** to monitor logs from a **Java application running on AWS EC2 (Ubuntu)**.

◆ 1. Overview of ELK Stack

The **ELK Stack** consists of:

- **Elasticsearch** → Stores and indexes logs.
 - **Logstash** → Processes and transforms logs before storing them in Elasticsearch.
 - **Kibana** → Provides visualization and analysis of logs.
 - **Filebeat** → Forwards logs from the application to Logstash.
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◆ 2. Infrastructure Setup

We are using **three EC2 Ubuntu machines**:

1. **ELK Server** → Hosts **Elasticsearch, Logstash, Kibana**.
 2. **Client Machine** → Hosts **Java application** and **Filebeat**.
 3. **Web Server (Optional)** → Hosts an **additional application** for testing logs.
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◆ 3. Step-by-Step Installation

◆ Step 1: Install & Configure Elasticsearch (ELK Server)

✓ 1.1 Install Java (Required for Elasticsearch & Logstash)

```
sudo apt update && sudo apt install openjdk-17-jre-headless -y
```

✓ 1.2 Install Elasticsearch

```
wget -qO - https://artifacts.elastic.co/GPG-KEY-elasticsearch | sudo apt-key add -
```

```
echo "deb https://artifacts.elastic.co/packages/7.x/apt stable main" | sudo tee  
/etc/apt/sources.list.d/elastic-7.x.list
```

```
sudo apt update
```

```
sudo apt install elasticsearch -y
```

✅ 1.3 Configure Elasticsearch

```
sudo vi /etc/elasticsearch/elasticsearch.yml
```

Modify:

```
network.host: 0.0.0.0
```

```
cluster.name: my-cluster
```

```
node.name: node-1
```

```
discovery.type: single-node
```

✅ 1.4 Start & Enable Elasticsearch

```
sudo systemctl start elasticsearch
```

```
sudo systemctl enable elasticsearch
```

```
sudo systemctl status elasticsearch
```

✅ 1.5 Verify Elasticsearch

```
curl -X GET "http://localhost:9200"
```

◆ Step 2: Install & Configure Logstash (ELK Server)

✅ 2.1 Install Logstash

```
sudo apt install logstash -y
```

✅ 2.2 Configure Logstash to Accept Logs

```
sudo vi /etc/logstash/conf.d/logstash.conf
```

Add:

```
input {  
  beats {  
    port => 5044  
  }  
}
```

```
filter {  
  grok {  
    match => { "message" => "%{TIMESTAMP_ISO8601:log_timestamp} %{LOGLEVEL:log_level}  
%{GREEDYDATA:log_message}" }  
  }  
}
```

```
output {
  elasticsearch {
    hosts => ["http://localhost:9200"]
    index => "logs-%{+YYYY.MM.dd}"
  }
  stdout { codec => rubydebug }
}
```

✅ 2.3 Start & Enable Logstash

```
sudo systemctl start logstash
```

```
sudo systemctl enable logstash
```

```
sudo systemctl status logstash
```

✅ 2.4 Allow Traffic on Port 5044

```
sudo ufw allow 5044/tcp
```

◆ Step 3: Install & Configure Kibana (ELK Server)

✅ 3.1 Install Kibana

```
sudo apt install kibana -y
```

✅ 3.2 Configure Kibana

```
sudo vi /etc/kibana/kibana.yml
```

Modify:

```
server.host: "0.0.0.0"
```

```
elasticsearch.hosts: ["http://localhost:9200"]
```

✅ 3.3 Start & Enable Kibana

```
sudo systemctl start kibana
```

```
sudo systemctl enable kibana
```

```
sudo systemctl status kibana
```

✅ 3.4 Allow Traffic on Port 5601

```
sudo ufw allow 5601/tcp
```

✅ 3.5 Access Kibana Dashboard

Open a browser and go to:

```
http://<ELK_Server_Public_IP>:5601
```

◆ Step 4: Install & Configure Filebeat (Client Machine)

✅ 4.1 Install Filebeat

```
wget -qO - https://artifacts.elastic.co/GPG-KEY-elasticsearch | sudo apt-key add -
```

```
echo "deb https://artifacts.elastic.co/packages/7.x/apt stable main" | sudo tee  
/etc/apt/sources.list.d/elastic-7.x.list
```

```
sudo apt update
```

```
sudo apt install filebeat -y
```

✅ 4.2 Configure Filebeat to Send Logs to Logstash

```
sudo vi /etc/filebeat/filebeat.yml
```

Modify:

```
filebeat.inputs:
```

```
- type: log
```

```
  enabled: true
```

```
  paths:
```

```
    - /home/ubuntu/Boardgame/target/app.log
```

```
output.logstash:
```

```
  hosts: ["<ELK_Server_Private_IP>:5044"]
```

✅ 4.3 Start & Enable Filebeat

```
sudo systemctl start filebeat
```

```
sudo systemctl enable filebeat
```

```
sudo systemctl status filebeat
```

✅ 4.4 Verify Filebeat is Sending Logs

sudo filebeat test output

◆ Step 5: Deploy Java Application & Generate Logs

✅ 5.1 Install Java (If Not Installed)

```
sudo apt install openjdk-17-jre-headless -y
```

✅ 5.2 Download & Run Sample Java App

```
wget https://repo1.maven.org/maven2/org/springframework/boot/spring-boot-sample-simple/1.4.2.RELEASE/spring-boot-sample-simple-1.4.2.RELEASE.jar -O app.jar
```

```
nohup java -jar app.jar > /home/ubuntu/Boardgame/target/app.log 2>&1 &
```

✅ 5.3 Verify Java Application is Running

✅ 5.4 Generate Logs for Testing

```
echo "Test log entry $(date)" >> /home/ubuntu/Boardgame/target/app.log
```

◆ Step 6: View & Analyze Logs in Kibana

✅ 6.1 Open Kibana Discover

1. Go to **Kibana** → **Discover**.
2. Select `log*` index.
3. Search for:

```
log.file.path: "/home/ubuntu/Boardgame/target/app.log"
```

4. View structured fields (`log_timestamp`, `log_level`, `log_message`).

✅ 6.2 Create Kibana Visualizations

1. **Pie Chart** → Log level distribution.
2. **Line Chart** → Logs over time.
3. **Data Table** → Structured log table.

✅ 6.3 Create a Kibana Dashboard

1. Go to **Kibana** → **Dashboard** → **Create Dashboard**.
2. Add **Pie Chart**, **Line Chart**, **Data Table**.

3. Save as "Java Application Log Monitoring".
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Conclusion

You have successfully:

- ✓ Installed **Elasticsearch, Logstash, Kibana, and Filebeat**
- ✓ Set up a **Java application to generate logs**
- ✓ Parsed logs into **structured fields using Grok**
- ✓ Created a **real-time Kibana dashboard for log monitoring** 🎉